Heat is not temperature

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Distinguish temperature from energy transferred because of a temperature difference.

SUCCESS CRITERIA - I can...

1. Compare temperature with thermal energy.
2. Identify the direction of spontaneous heat transfer.
3. Explain why amount and material matter.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A simplified model requires 15 J to escape, but an object has only 12 J and no later energy input. Does the model predict escape? Explain.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $Q = mc\,\Delta T$.
For heating without a phase change: Q is transferred heat, m is mass, c is specific heat. Heat and temperature are different quantities.

Worked reasoning: A cup and a bathtub can both be at 40°C. Equal temperature does not imply equal internal energy: amount and type of material also matter.

Your turn: Draw equal temperature indicators but different amounts of water. Predict heat-flow direction if the water samples are brought into thermal contact at the same temperature.

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

Marzano check - circle: ① ② ③

03 Read and interpret the model

A cup and a bathtub contain the same substance at 40°C. Are their temperatures equal? Must their total internal energies be equal?





Cup: 40°C

Tub: 40°C

Each outline represents the labeled sample or container.

Marzano check - circle: ① ② ③

04 Apply the physics

A metal spoon at 60°C is placed in water at 20°C. Label both samples and draw the initial energy-transfer arrow. Predict which temperature rises and which falls.





Spoon: 60°C

Water: 20°C

Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

A student says “cold flows into a hot drink.” Rewrite the explanation in terms of energy transfer.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A wheel rolls down a ramp, losing 24 J gravitational energy. Its center-of-mass kinetic energy gain is 20 J with negligible losses. What energy term is missing?
Show the energy changes and transfers; label units.

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

07 Exit check

Label a thermometer reading and an energy-transfer arrow on the model. Explain one difference between temperature and heat, including the usual units for each.

Initial T / phase: _____ Final T / phase: _____



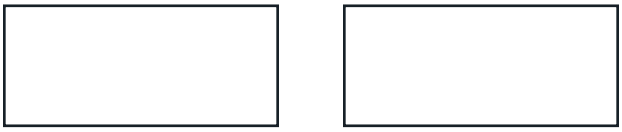
Add an energy arrow; label mass, c or L as needed.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A small metal spoon is hotter than a large bath. Can temperature alone tell which contains more internal energy? Explain what else you would need.

Sample A Sample B



Label temperatures and draw energy-transfer arrows.

Marzano check - circle: 1 2 3